

TØ 1 – Mikroøkonomi 2 – 2024

Problems for Tutorial

Please try to solve all exercises on your own BEFORE class. Please try to get as far as possible with each problem. If you cannot solve an exercise, try to sum up why you got stuck, so you can check if this got clarified in class.

Problem 1

Ask and Benedikte share an apartment and a love for plants. They have different houseplants that they want to display around their apartment. Each of them has two big potted plants and needs to determine how many to place in their own room and how many to position in the common areas. Suppose that the decision is made privately and that once the plants are in their place, due to their size, they cannot be moved.

Let x_A and x_B be the number of plants that Ask and Benedikte, respectively, decide to place in their own room. Then $x_L = 4 - x_A - x_B$ is the number of plants in the living room. Let the utility function of Anna and Bertram be $u_A(x_a, x_b) = x_A(1.5 + x_L)$ and $u_B(x_a, x_b) = x_B(1.5 + x_L)$, respectively.

- (a) Which are the available actions for each of the roommates?
- (b) Represent the game as a matrix.
- (c) What is your prediction? Explain briefly.

a.

Both Ask and Benedikt can have either 0, 1 or 2 paintings in the living room
Mathematically, for each player $i = (A, B)$, available actions are $\{0, 1, 2\}$

b.

Where the number represents number of paintings each person will store in their own room.
An example of the calculations is (1,0), i.e., when Ask stores 1 painting in her own room and Benedikt 0:

$$u_A(x_a, x_b) = x_A(1.5 + x_L) = 1(1.5 + 3) = 4.5$$

$$u_B(x_a, x_b) = x_B(1.5 + x_L) = 0(1.5 + 3) = 0$$

Another example: (1,2):

$$u_A(x_a, x_b) = x_A(1.5 + x_L) = 1(1.5 + 1) = 2.5$$

$$u_B(x_a, x_b) = x_B(1.5 + x_L) = 2(1.5 + 1) = 5$$

A \ B	0	1	2
0	0, 0	0, 4.5	0, 7
1	4.5, 0	3.5, 3.5	2.5, 5
2	7, 0	5, 2.5	3, 3

c.

Even though it would be better for Ask and Benedikt to store one painting each in the living room, they will not. As an example:

Suppose Benedikt is trying to decide whether he should place a painting in the living room or not.

If Ask places a painting in the living room (1: the first row), he could either: receive 3.5 in utility or receive 5.0 in utility

If Ask does not place a painting in the living room (2: the second row) he could either: receive 2.5 or 3.0 in utility.

Suggesting that no matter what Ask chooses, he should place both paintings in his own room even though the utility would get higher for both parties if they agreed on place one painting each in the living room.

This implies that $x_i = 2$ is a strictly dominant strategy for both players

Problem 2

Two people enter a bus. Two adjacent seats are free. Each person must decide whether to sit or stand. Sitting alone is more comfortable than sitting next to the other person, which is more comfortable than standing.

- (a) Suppose that each person only cares about her own comfort. Model the situation as a strategic game. Is this game the *Prisoner's Dilemma*?
- (b) Suppose that each person is altruistic, ranking the outcomes according to the other person's comfort, but, out of politeness, prefers to stand than to sit if the other person stands. Model the situation as a strategic game.
- (c) What are your predictions for the games in part (a) and (b), respectively?

a.

It is not the prisoner's dilemma because no matter what the other person chooses, it is more attractive to sit down than to stand up, and there is not a scenario where the two parts "destroy" the probability of reaching the equilibrium with the highest payoff

A handwritten payoff matrix on grid paper. The columns are labeled 'stand' and 'sit' at the top. The rows are labeled 'stand' and 'sit' on the left. The payoffs are written in the cells: (stand, stand) is 0,0; (stand, sit) is 0,2; (sit, stand) is 2,0; and (sit, sit) is 1,1.

	stand	sit
stand	0,0	0,2
sit	2,0	1,1

And prisoner's dilemma is:

		PRISONER 2	
		Confess	Lie
PRISONER 1	Confess	-8 , -8	0 , -10
	Lie	-10 , 0	-1 , -1

b.

A brief explanation: Suppose that player 1 stands, i.e., we are in the first row. Then player 2 prefers to stand as well out of politeness pictured by a negative utility by sitting down when player is standing.

	Stand	Sit
Stand	0, 0	2, -1
Sit	-1, 2	1, 1

According to IESDS, we get

Part a: (sit, sit)

Part b: (stand, stand)

1\2	Step	S.7
Step	0,0	0,2
S.7	2,0	1,1

1\2	Step	S.7
Step	0,0	2,-1
S.7	-1,2	1,1

Problem 3

Some studies suggest that tobacco advertising does not affect the total number of packs sold, but rather shifts smokers among the different brands. Suppose the total annual demand for cigarettes is 1,000,000,000 packs and that the market share of a company depends on how much it spends on advertising relative to what its rival spends. Suppose there are only two producers: Philip Morris (PM) and British American Tobacco (BAT). Let a_i be the advertising expenditures of player i , $i = \text{PM}, \text{BAT}$. The market share of player i equals

$$\frac{a_i}{a_i + a_j} \quad i \neq j$$

so that the total number of packs sold by i is

$$1,000,000,000 \cdot \frac{a_i}{a_i + a_j}.$$

A firm's profit per pack is 10 cents. Suppose that firms can choose one of three levels of advertising expenditure: \$5 million, \$10 million, and \$15 million.

- (a) Model this situation as a normal form game where players' payoffs correspond to their profits. What is your prediction on the two firms' level of advertising spending?
- (b) Now suppose that the government introduces regulations that make it infeasible for the two cigarette companies to spend \$15 million on advertising. How does your prediction change?

a.

First, I simplify the total number of packs sold:

$$1,000 \frac{a_i}{a_i + a_j}$$

The profit per pack is 10 cents. Thus, the profit of player i can be written as total number of packs times sold by i times profit per pack minus the advertising expenditure:

$$\left(1,000 \frac{a_i}{a_i + a_j} \right) \cdot 0.1 - a_i$$

I can now draw the matrix

As an example, if player 1 spends \$5 million on advertising and player 2 spends \$10 million on advertising, the profit of player 1 will be:

$$\pi_{5,10} = \left(1000 \cdot \frac{a_i}{a_i + a_j} \right) \cdot 0.1 - a_i = \left(1000 \cdot \frac{5}{5 + 10} \right) \cdot 0.1 - 5 = 28.33$$

And for player 2

$$\pi_{10,5} = \left(1000 \cdot \frac{a_i}{a_i + a_j} \right) \cdot 0.1 - a_i = \left(1000 \cdot \frac{10}{5 + 10} \right) \cdot 0.1 - 10 = 56.67$$

And so on:

BAT

		5	10	15
PM	5	45, 45	28.33, 56.67	20, 60
	10	56.67, 28.33	40, 40	30, 45
	15	60, 20	45, 30	35, 35

I use IESDS to predict the outcome: Both companies will spend \$15 million on advertising

BAT

		5	10	15	
PM	5	45, 45	28.33, 56.67	20, 60	1.
	10	56.67, 28.33	40, 40	30, 45	2.
	15	60, 20	45, 30	35, 35	
		3.	4.		

b.

This implies that we remove the third column and third row. Now, the two players will each spend \$10 million on advertising according to IESDS

Problem 4

Consider only pure strategies in the following games:

Game 1	1 \ 2	b_1	b_2	b_3
	a_1	2,0	1,1	4,2
	a_2	1,4	1,2	2,3
	a_3	1,3	0,2	3,0

Game 2	1 \ 2	b_1	b_2	b_3
	a_1	4,5	1,6	5,6
	a_2	3,5	2,5	5,4
	a_3	2,5	2,0	7,0

- (a) Which strategies survive *iterated elimination of strictly dominated strategies* (IESDS) and which survive *iterated elimination of weakly dominated strategies* (IEWDS) in Game 1?
- (b) Which strategies survive *iterated elimination of strictly dominated strategies* (IESDS) and which survive *iterated elimination of weakly dominated strategies* (IEWDS) in Game 2?

In game 1, the strategy $(a_1, b_3) = (4, 2)$ survives IESDS and is therefore the equilibrium. Because it survived IESDS, it will also survive IEWDS because

$$x > y \Rightarrow x \succcurlyeq y$$

	b_1	b_2	b_3
a_1	2,0	1,1	4,2
a_2	1,4	1,2	2,3
a_3	1,3	0,2	3,0

4. 2.

b.

No strategies survive IESDS, i.e., no strategy strictly dominates another. However, one strategy survives IEWDS: (a_2, b_2)

So, if a strategy survives IESDS it also survives IEWDS but not the other way around

	b_1	b_2	b_3	
a_1	4,5	1,6	5,6	4.
a_2	3,5	2,5	5,4	
a_3	2,5	2,0	7,0	2.
	3.		1.	

Problem 5

Three researchers ($i = 1, 2, 3$) are working on a joint project. The amount of time researcher i spends on the project is denoted $x_i \geq 0$. The quality q of the research output depends on x_1, x_2 , and x_3 as follows:

$$q(x_1, x_2, x_3) = 2x_1 + 2x_2 + x_3 - x_1x_2 - x_1x_3.$$

Spending time on the project is costly for the researchers. The cost function for each of them is:

$$c_i(x_i) = (x_i)^2.$$

The utility for each researcher is equal to the quality of the output minus her cost:

$$u_i(x_1, x_2, x_3) = q(x_1, x_2, x_3) - c_i(x_i).$$

- Find the best response functions for the three researchers. I.e., for each i , find the optimal amount of time to spend on the project given the time spent by the other two.
- Suppose the researchers simultaneously and independently decide how much time to spend on the project. Find the Nash equilibrium of this game.

a.

Utility of researcher 1, 2 and 3, respectively, is

$$u_1(x_1, x_2, x_3) = 2x_1 - x_1x_2 - x_1x_3 - x_1^2$$

$$u_2(x_1, x_2, x_3) = 2x_2 - x_1x_2 - x_2^2$$

$$x_3 - x_1x_3 - x_3^2$$

Thus, the FOCs of the researchers are

$$2 - x_2 - x_3 - 2x_1 = 0$$

$$2x_2 - x_1 - 2x_2 = 0$$

$$1 - x_1 - 2x_3 = 0$$

BR are found by isolating for x_i in the FOCs:

$$2 - x_2 - x_3 - 2x_1 = 0 \Leftrightarrow x_1 = \frac{2 - x_2 - x_3}{2}$$

$$x_2 = 1 - \frac{1}{2}x_1$$

$$x_3 = \frac{1 - x_2}{2}$$

b.

The BRs are three equations with three unknowns. Solving this gives

$$x_1^* = \frac{1}{2}$$

$$x_2^* = \frac{3}{4}$$

$$x_3^* = \frac{1}{4}$$